Chapter 7. The Bootstrap

Sampling distributions, bootstrap estimates of standard error and bias, resampling schemes

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7.1 The plug-in principle

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ and $\hat{\theta} = T(X_1, \dots, X_n)$ estimate $\theta = \theta(F)$. The **sampling distribution** G_F of $\hat{\theta}$, and with it the bias, standard error and quantiles, depends on the unknown F. If we could draw new samples from F we would simply simulate. The bootstrap replaces F by an estimate $\hat{F}$ and simulates from $\hat{F}$:

$$\text{real world: } F \to \mathbb{X}_n \to \hat{\theta}, \qquad \text{bootstrap world: } \hat{F} \to \mathbb{X}_n^* \to \hat{\theta}^*.$$

The distribution of $\hat{\theta}^*$ given the data, $G^* = G_{\hat{F}}$, is the **bootstrap distribution**; it estimates G_F.

Nonparametric bootstrap. $\hat{F} = F_n$, the empirical distribution (mass $1/n$ at each observation). Sampling from F_n means drawing n observations *with replacement* from the data.

1. Draw a bootstrap sample $X_1^*, \dots, X_n^*$ with replacement from $\{X_1, \dots, X_n\}$ and compute $\hat{\theta}^* = T(X_1^*, \dots, X_n^*)$.
2. Repeat B times to get $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$; their empirical distribution approximates G^*.

There are two errors: $F_n \neq F$ (statistical, of order $n^{-1/2}$, the important one) and $B < \infty$ (Monte Carlo, of order $B^{-1/2}$, reducible at will).

7.2 Bootstrap estimates of standard error and bias

Replacing F by F_n in $\text{se}_F(\hat{\theta})$ and $\text{bias}_F(\hat{\theta}) = E_F\hat{\theta} - \theta(F)$ gives

$$\widehat{\text{se}}_{\text{Boot}} = \Big(\frac{1}{B-1}\sum_b (\hat{\theta}^{*b} - \bar{\theta}^*)^2\Big)^{1/2}, \qquad \widehat{\text{bias}}_{\text{Boot}} = \bar{\theta}^* - \hat{\theta}, \qquad \bar{\theta}^* = \frac{1}{B}\sum_b \hat{\theta}^{*b}.$$

Note that the role of the true $\theta(F)$ is played by $\theta(F_n)$, which is $\hat{\theta}$ for plug-in statistics.

Example 7.1 (An exact bootstrap distribution)

Data $\{1, 4, 9\}$, $\hat{\theta}$ = median = 4. Among the $3^3 = 27$ equally likely resamples, the median equals the smallest value 1 iff at least two draws are 1: probability $(3 \cdot 2 + 1)/27 = 7/27$. Likewise it equals 9 with probability 7/27, and 4 with probability 13/27. Hence

$$E^*\hat{\theta}^* = \frac{7 \cdot 1 + 13 \cdot 4 + 7 \cdot 9}{27} = 4.52, \qquad \text{se}^* = \Big(\frac{7 + 13 \cdot 16 + 7 \cdot 81}{27} - 4.52^2\Big)^{1/2} = 2.92,$$

and the bootstrap bias estimate is 0.52. With B random resamples these values are only approximated. Here they can be computed exactly because n is tiny.

When the bootstrap works

For smooth statistics (means, smooth functions of means, many M- and U-statistics) the bootstrap distribution of $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$ consistently estimates that of $\sqrt{n}(\hat{\theta} - \theta)$. It can fail for non-smooth targets at the boundary of the parameter space, e.g. $\hat{\theta} = \max_i X_i$ for Uniform$(0, \theta)$, where $P^*(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/n)^n \to 1 - e^{-1}$, whereas $P(\hat{\theta} = \theta) = 0$.

7.3 Parametric bootstrap

If $F = F_\eta$ belongs to a parametric family, estimate $\hat{\eta}$ (e.g. by maximum likelihood) and draw bootstrap samples from $F_{\hat{\eta}}$ instead of F_n. When the model is right this is more accurate than the nonparametric bootstrap, and it is useful when the sampling distribution of $\hat{\theta}$ is analytically intractable (e.g. a nonlinear functional of a Weibull or gamma fit). When the model is wrong it can be badly biased.

7.4 Resampling designs

The guiding rule is: *resample the way the data were generated.*

- **Several samples.** For independent samples from $F_1, \ldots, F_k$, resample within each sample separately, keeping the sample sizes $n_1, \ldots, n_k$.
- **Bivariate data, random design.** If $(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} F$, resample the pairs (**case** or pairs bootstrap). This keeps the dependence between X and Y and is valid even if a regression model is wrong.
- **Regression, fixed design.** If $Y_i = \alpha + \beta x_i + \epsilon_i$ with fixed x_i and i.i.d. errors, resample residuals: $Y_i^* = \hat{\alpha} + \hat{\beta} x_i + r_i^*$, with r_i^* drawn from the (leverage-adjusted) residuals. This is more efficient when the model is right but relies on it, including homoscedasticity.

Contrast with permutation tests: the bootstrap samples *with* replacement and keeps the structure of the alternative (e.g. the pairs (X_i, Y_i) stay together). A permutation test samples *without* replacement and deliberately destroys the association, to produce the null distribution.